

Qatar-4b

R-2681

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-4_180929 · created 2026-08-30T23:41:29+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458390.61253 +/- 0.00056 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.177 +/- 0.051
Transit depth (Rp/Rs)^2	3.14 +/- 1.8 [%]
Orbital Inclination (inc)	87.54 +/- 2.0
Airmass coefficient 1 (a1)	0.99 +/- 0.019
Airmass coefficient 2 (a2)	0.002 +/- 0.037
Scatter in the residuals of the lightcurve fit is	1.96 %
Best Comparison Star	#3 - [332, 343]
Optimal Method	PSF photometry
Transit Duration (day)	0.0582 +/- 0.004

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.177 ± 0.051 vs 0.138 (deviation 0.76σ)
Duration fit vs published	0.0582 d vs 0.0892 d (deviation 7.74σ)
Mid-time O–C	0.9 min
Depth SNR	3.5
Residual scatter	1.96 %

Light curve

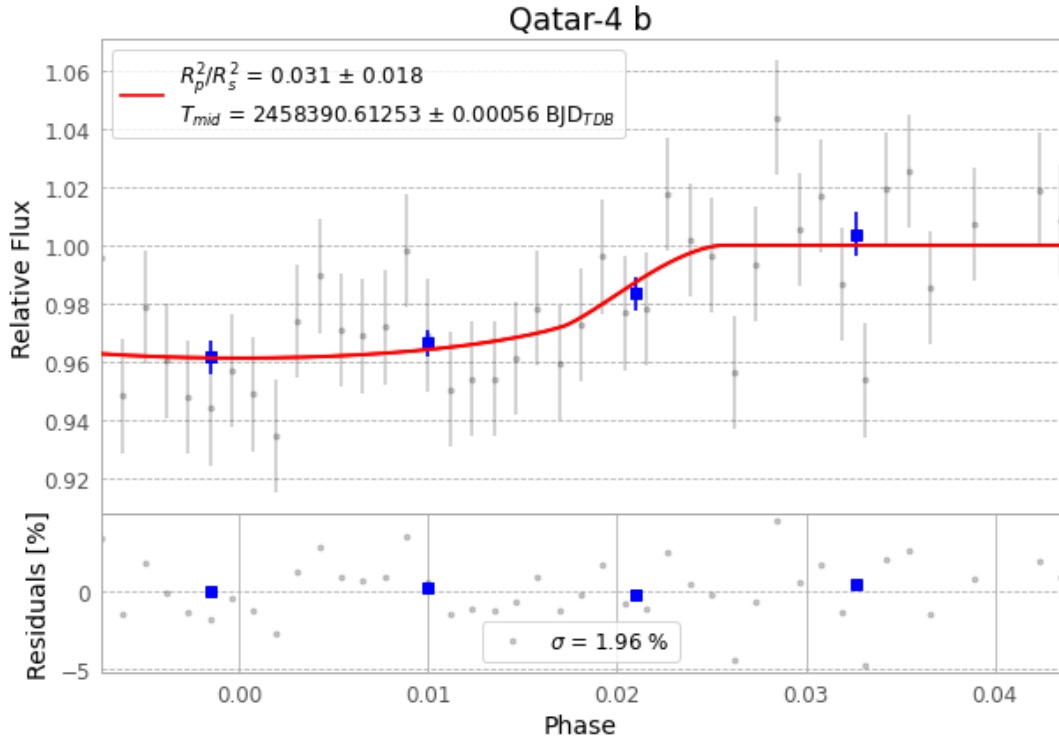


Figure 1 — FinalLightCurve_Qatar-4 b_2018-09-28.png (SHA-256 9f97542c49e680db...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [366, 175], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 3d3ea3599106c55b...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

45 FITS frames (30 MB), Qatar-4180929021512.FITS → Qatar-4180929042712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-4-180929.log (1bb42d4149bf...), FinalLightCurve_Qatar-4 b_2018-09-28.png (9f97542c49e6...), FinalLightCurve_Qatar-4 b_2018-09-28.csv (de4de9c40396...)

Compute resources

Wall time 143.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-30 23:41Z.